KEDGE Business School EBP1 Quantitative Methods 1

# Worksheet 2 — Measures of central tendency

**Exercise statements**

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## 1 Core applications

Master the essential definitions and calculations from the course.

*Core application — 25 min*

### 1.1 Exercise 1 – Choosing and calculating the appropriate measure of centre

1. Basket values are $18, 20, 20, 22, 25, 95$ euros. Calculate mode, median and arithmetic mean. Which measure best represents a typical customer? Why can the mean still matter?
2. Three grades are 14, 11 and 16, weighted by 30%, 30% and 40%. Calculate the weighted mean.
3. Revenue changes by $+5\%$, $-2\%$ and $+12\%$. Calculate the appropriate average annual growth rate.
4. A vehicle travels 120 km at 60 km/h and then 120 km at 90 km/h. Calculate the appropriate average speed.

*Core application — 10 min*

### 1.2 Exercise 3 – Which indicator should be used?

For each situation, choose the most appropriate indicator and justify in one sentence: most frequently sold product category; typical salary with a few very highly paid executives; average price when purchased quantities differ; average multi-year growth; average speed over equal distances; centre of a roughly symmetric numerical distribution without extreme values.

*Core application — 15 min*

### 1.3 Extra 1 — Robust centre for delivery performance

Delivery times (minutes) for 12 orders are: 18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 65.

1. Calculate the mode, median and mean.
2. Recalculate the mean without the 65-minute delivery.
3. Recommend the principal indicator for a customer-facing dashboard and justify your choice.

**Method hint:**

Compare the indicators before and after removing the unusual observation; do not delete it without explanation.

## 2 Assessment and exam practice

Practise independently with assessment and final-exam style questions.

*Assessment / exam practice — 30 min*

### 2.1 Exercise 2 – Grouped distribution

| Class | $[10, 14[$ | $[14, 18[$ | $[18, 26[$ | $[26, 30[$ | $[30, 35[$ | $[35, 40[$ | $[40, 45[$ | $[45, 50[$ | $[50, 55[$ |
|---|---|---|---|---|---|---|---|---|---|
| $n_i$ | 4 | 12 | 32 | 32 | 60 | 35 | 25 | 5 | 5 |

1. Identify the modal class, taking unequal widths into account.
2. Calculate the median by linear interpolation.
3. Calculate $Q_1$ and $Q_3$ by interpolation.
4. Calculate the approximate mean using class midpoints.
5. Find the maximum price such that about 60% of customers can afford at least that price.

*Assessment / exam practice — 20 min*

### 2.2 Extra 2 — Integrated centre-selection case

A retailer reports three indicators:

| Question | Data |
|---|---|
| Average basket value | 120 baskets: €3,600 total |
| Typical annual growth | +12%, −8%, +15%, +5% |
| Average speed | Two equal 90-km legs at 60 and 90 km/h |

1. Choose and calculate the appropriate mean for each row.
2. Explain why using the arithmetic mean in all three cases is incorrect.
3. Write a short managerial conclusion.

**Method hint:**

Match additive totals, compounded factors and equal-distance rates with arithmetic, geometric and harmonic means.